

Dr Gopal explains, “This is a prime example where open source can play a role. I believe that anything that is being created for the public using public funds should be open source. It should be done as a collaborative project where multiple parties can work together. We have tried to create a model that is open source and have also created the organisational construct. So, different parties can contribute code and create the governance model for an open source project. This takes a lot of effort and, very often, it is really difficult to maintain and manage because there are always going to be conflicts.”

Collaborate in the middle layer: He adds, “What we have done is disaggregate the deployment of smart city systems. Previously, a city would go to a vendor and the latter would build a vertically integrated system, but we now realise that this is not the way forward. We have to split these systems into horizontal layers.”

At the bottom, there are the sensors from different vendors. These can be mixed and matched as cities now have the ability to be vendor-agnostic. Sensors can be purchased, let’s say, from Samsung and then something can be bought from Philips, and the companies can compete against each other at that level. The idea really is innovation below at the data collection level and then innovation at the application level. “Create applications and services on top. Collaborate in the middle to maximise the availability and involve more players,” Dr Gopal adds.

Bringing cross-industry solutions together: The smart city construct is similar all across the globe, and the focus is on how cross-industry solutions can be brought together and automated. “For example, how can we support factories of the future and bring in these technologies together, helping in the automation, in the right kind of connectivity, and in bringing insights so that the operations efficiency can be improved,” Gnanapriya opines.

Then there are the surveillance, safety and security solutions. As an example, in the oil and gas industries, there seem to be a lot of opportunities for these solutions, but there are challenges in accessing

or processing them in real-time. “It is important to think about how to bring all the partner ecosystems together in an operations platform that can help in taking solutions forward,” she adds.

Work towards forming a global registry: Gajare says, “We have a few open source smart city platforms (for Pune, Jaipur and Navi Mumbai), and we are also trying to do something for the digital freight corridor operation now. But we need a registry for digital public goods, a place where all this data can be exchanged at par with the right security levels and context.”

“Having a global registry for smart cities, or even just for the 6000 odd cities in India, would be nice, and that is solely missing as one of the key agenda items.”

Improve edge computing capabilities: There are two types of edge — user edge and service provider edge. And the user edge may be either the microcontroller based or gateway systems. There is a focus on creating network infrastructure that may be a private network, especially to enable edge computing. Experts warn that you may be forced to enhance edge computing in future solutions.

Chaudhary says, “Smart factories and rural connectivity need a private network in a lot of cases. We are trying to build an open network operating system that needs three components — the radio, transport and the core. We can create a concrete edge upfront core network for 5G using open source. It is all about collaboration, about more competitors coming and collaborating.”

Open source drives standards: Dr Gopal reiterates the relationship between standards and open source: “The way standards get developed is by creating open source reference implementations. There is a kind of cycle between formal standards and implementations. And in many ways, open source drives those standards.”

A task force in the Bureau of Indian Standards is looking at the feasibility of 5G and open source for 5G, as well as multiple segments, such as the access and applications space, to check on

the maturity of open source. “But the objective really is to take it forward and go from a paper study to actually do a set of reference implementations in each of these areas,” says Dr Gopal.

Developing the culture of contributions: India is known for its software programming, but when it comes to open source, the country is still in its infancy. Gajare feels the time has come for creating something similar for the whole 5G ecosystem, where everyone who is a student, academician or research scholar can contribute. India has the mental muscle and now it’s just a matter of working with the necessary frameworks that align to solve something tangible.

“We have huge talent in India. Getting associated with any open source project of interest will help. Students can start with bug fixes, and then contribute to features. There are a variety of opportunities available at any layer,” says Gnanapriya.

Experts believe that we must develop the culture of contributions rather than looking at reaping immediate benefits. “I think we need to really grow our thought process and nurture this culture, also by busting a lot of myths around it. If you are contributing something to open source, it is not your intellectual property. Let’s build this culture of contributions,” Gangey concludes. 

Panelists

- Dr Alok Nath De, CTO, Samsung India (moderator)
- Dr Inder S. Gopal, CEO, Indian Urban Data Exchange
- Manish Gangey, SVP and Head, R&D, Airtel
- Abhijit Chaudhary, founder and CEO, Nirala Networks
- Gnanapriya Chidambaramathan, AVP and senior principal architect, Infosys
- Subodh Gajare, lead architect (5G and IoT security), Cisco R&D

 **By: Abhinaya Kuzhanthaivel**

The author works as an assistant editor at EFY.